Stop-Aware Reward and Extended Training for PointGoal Navigation

From Habitat Simulation to Real-Robot Deployment on a Low-Cost Platform

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Abstract — PPO-trained PointGoal navigation policies in Habitat show two typical failures: near-miss (reaching the goal but not stopping) and lost (navigation failure). We propose Stop-Aware Reward — episode-end reward shaping that fixes stop-related failures (bad_stop −80%, cross-seed SR std 0.063→0.011) — and show that extending training from 1e7 to 5e7 steps yields the largest gain (SR 0.890→0.945, SPL 0.731→0.820), far exceeding four PPO hyperparameter ablations. The final policy was deployed end-to-end on a WHEELTEC S100 robot (Jetson Orin Nano + Astra S RGBD, ROS1) and reached goals in a real office environment.

1 · Motivation & Problem

- **PointGoal navigation.** reach a relative goal (polar distance, angle) without a pre-built map; PPO on Habitat + HM3D is the standard baseline.
- **No explicit stop signal.** the distance-shrinking reward r = Δd − 0.01 never rewards stopping, so agents learn stop timing implicitly.
- **Two stop-related failures:** near-miss (reaches the goal but keeps moving) and bad_stop (stops far from the goal); lost = pure navigation failure.
- **Failure analysis of 3-seed baseline (600 episodes):** 93 failures → 58 lost (62.4%), 30 near-miss (32.3%), 5 bad_stop (5.4%).

Failure type	Definition	Count	Share
Lost	DTG > 1.0 m	58	62.4%
Near-miss	0.2 < DTG ≤ 0.35 m	30	32.3%
Bad stop	0.35 < DTG ≤ 1.0 m	5	5.4%

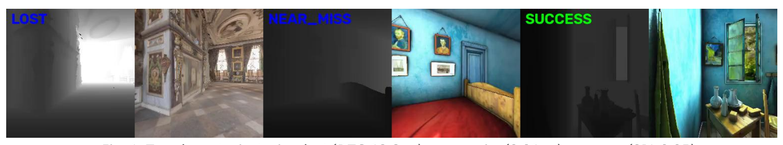


Fig. 1 Top-down trajectories: lost (DTG 12.8 m) · near-miss (0.21 m) · success (SPL 0.95)

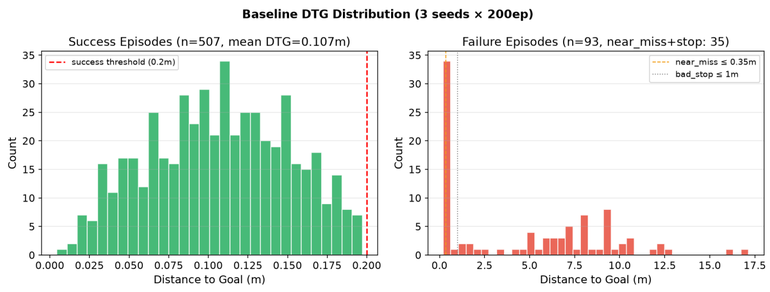


Fig. 2 Baseline DTG distribution — near-miss failures sit just 0.01 m outside the 0.2 m success radius

2 · Method

- **Baseline.** ResNet18 (RGB+Depth 256², HFOV 90°) + GRU(512) → 4 actions {stop, forward 0.25 m, turn ±10°}; reward r = Δd − 0.01; 500-step timeout.
- **Stop-Aware Reward.** at episode end (stop called), inject a one-shot signal; +2.0 for a correct stop (DTG ≤ 0.2 m), −1.0 otherwise; no signal on timeout episodes.
- **~20 lines, zero framework changes.** implemented as a Python monkey-patch of DistanceToGoalReward — compatible with any distance-based PointNav reward.

R_stop = +2.0 if DTG ≤ 0.2 m (correct stop)
R_stop = −1.0 if DTG > 0.2 m (bad stop)
One-shot signal at episode termination only — ≈20% of a successful episode's total reward.

3 · Training Setup

- **Data.** HM3D — 800 training / 20 validation scenes; 200 episodes per seed; seeds 100 / 200 / 300.
- **PPO.** clip ε=0.2, GAE λ=0.95, lr 2.5e-4, entropy 0.005, 10 parallel envs × 128 steps/rollout.
- **Compute.** single NVIDIA RTX 4090D; 1e7 steps ≈ 7 h, 5e7 steps ≈ 35 h.

Metrics — SR: stopped with DTG ≤ 0.2 m · SPL: success weighted by path length · DTG: distance to goal at termination. Failure classes — lost: DTG > 1.0 m · near-miss: 0.2–0.35 m · bad_stop: 0.35–1.0 m.

Key contributions

1. Stop-Aware Reward — lightweight episode-end shaping; bad_stop −80%, training consistency ×3 (std 0.063→0.011).
2. Systematic ablations (10 runs): more training steps ≫ PPO hyperparameter tuning (all 4 variants negative).
3. First end-to-end sim-to-real PointNav on a low-cost differential robot — SR 0.945 policy reaches goals in a real office.

Discussion

- Data coverage is the bottleneck: 5e7 steps (~35 h on one 4090D) outperform all four hyperparameter ablations combined — consistent with Habitat's own 75–250M-step PointNav training.
- Negative results matter: default Habitat PPO hyperparameters are already near-optimal for HM3D PointNav; reporting failed tuning directions saves the community GPU time.

4 · Results I — Stop-Aware Reward

Metric	baseline (1e7)	stop_aware (1e7)	Δ
Val SR	0.845	0.890	+4.5 pp
Val SPL	0.687	0.731	+4.4 pp
Val DTG	0.843 m	0.780 m	−7.5%
Cross-seed SR std	0.063	0.011	−82.5%
Bad stops / 600 ep	5	1	−80%

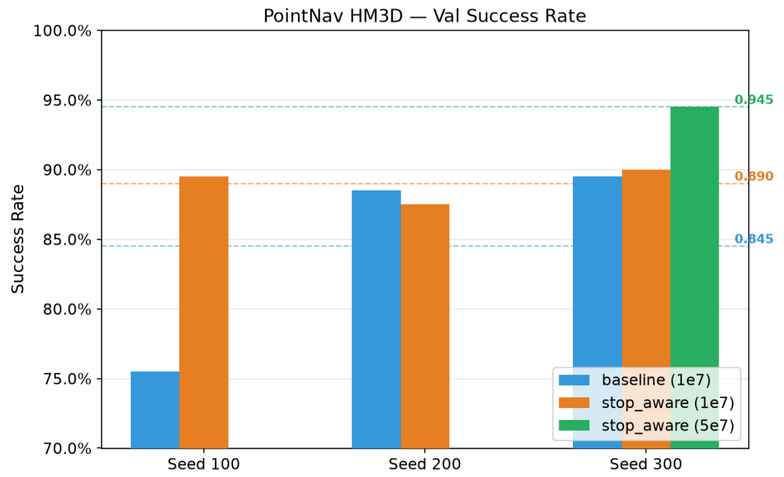


Fig. 3 Val success rate — every seed improves with stop-aware; 5e7 reaches 0.945

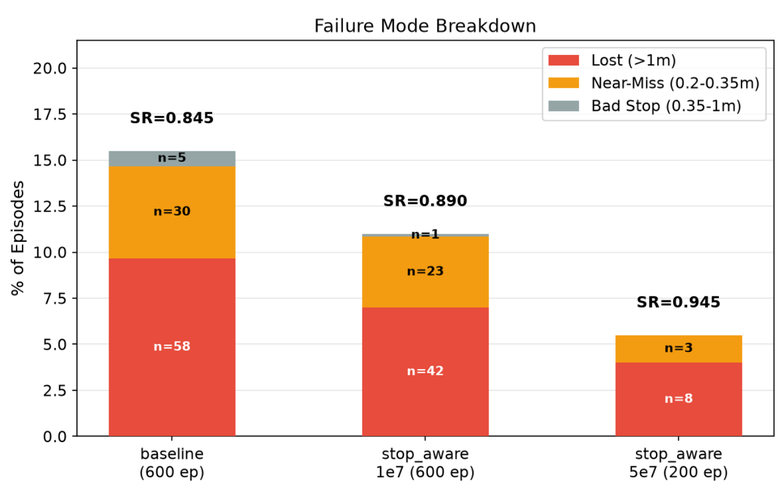


Fig. 4 Failure modes — stop-aware fixes stop-related failures; 5e7 further cuts lost episodes

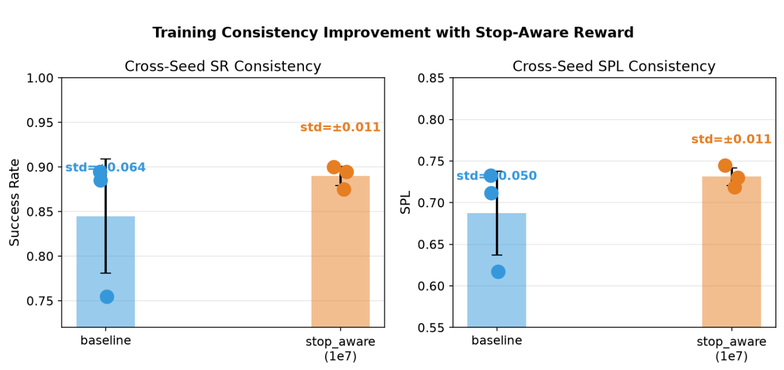


Fig. 5 Cross-seed consistency — explicit stop semantics stabilise training (SR std 0.063 → 0.011)

5 · Results II — Extended Training

Config	Val SR	Val SPL	Val DTG	Lost
1e7 steps	0.900	0.745	0.842 m	15
5e7 steps	0.945	0.820	0.515 m	8
Δ	+4.5 pp	+10.1%	−38.8%	−47%

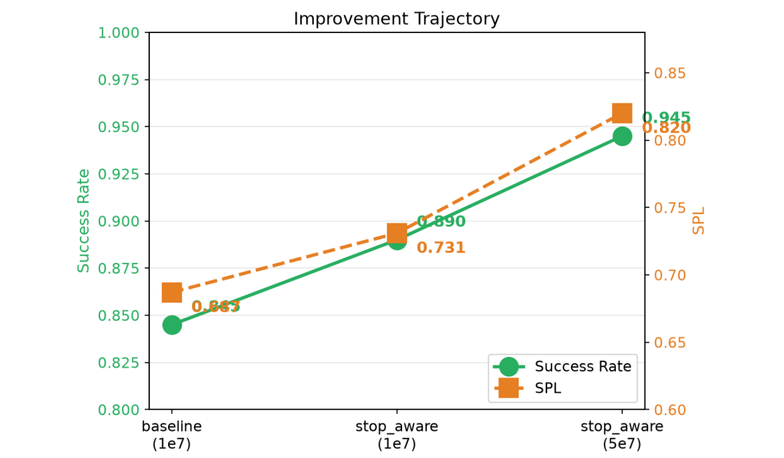


Fig. 6 Improvement trajectory — steps are the highest-leverage change; curves keep improving with no saturation

6 · Negative Results — PPO Hyperparameters

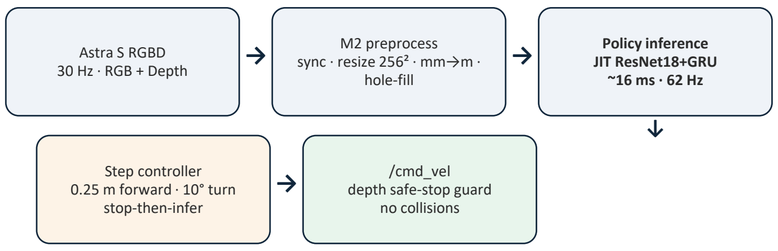
- **All four variants failed to help** (seed 100, 1e7 steps): linear LR decay — superficial; normalized advantage + τ=0.99 — collapsed; τ=0.99 alone — harmful (value loss ×3); advantage normalization — negative.
- **Take-away.** Habitat-Lab's default PPO is already well-tuned for PointNav; at 1e7 steps, scene coverage dominates and hyperparameter tuning has low signal-to-noise. Direction terminated.

7 · Sim-to-Real Deployment

- **Direct transfer.** trained in Habitat, weights loaded onto the robot as-is; sensor preprocessing + a discrete-action layer bridge the sim-real gap — no Gazebo, no external planner.

Component	Specification
Compute	Jetson Orin Nano (6-core ARM, 7.3 GB RAM)
OS / ROS	Ubuntu 20.04 · ROS1 Noetic
RGBD camera	Orbbec Astra S, 640×480, HFOV ≈57.6°
Localization	Wheel odometry + IMU EKF (robot_pose_ekf)
Drive	Differential, max 0.22 m/s
Inference	Python 3.8 · PyTorch Jetson build, JIT export

Deployment pipeline (step mode, ~62 Hz inference):



Model	Sim SR	Real-world test
baseline (1e7)	0.895	✓ Goal reached
stop_aware 5e7	0.945	✓ Goal reached

- **FOV gap.** trained at 90° HFOV vs ~57.6° on Astra S → reduced edge perception.
- **Weak texture.** RGB topic is actually IR grayscale; a depth-only JIT model is provided for low-texture environments.
- **Depth holes.** structured-light voids (~60.7% valid centre pixels) mitigated by median hole-filling.
- **Safety.** depth-based stop when centre-region min depth < threshold (0.2 m floor / 0.05 m hanging).

8 · Extended-Training Dynamics

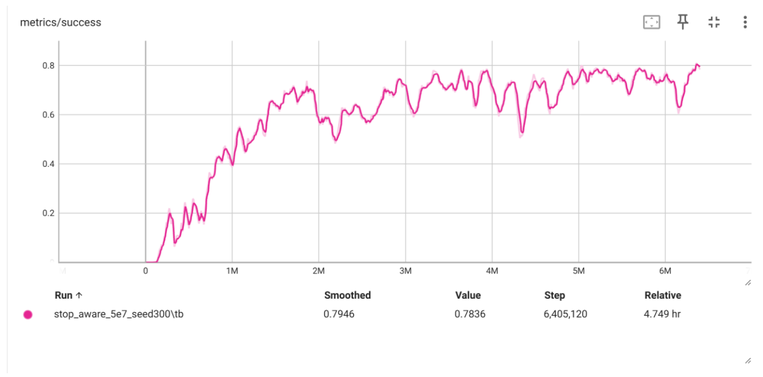


Fig. 7 TensorBoard success curve — no saturation at 5e7 (grad_norm 15.97→7.47)

- **No saturation.** the success curve is still climbing at 5e7 steps — 1e7 steps is far below the performance plateau.

SR	SPL	DTG	bad_stop
0.945	0.820	0.515 m	5 → 1
+10.0 pp	+13.3 pp	−39%	−80%

9 · Conclusion & Future Work

- **Stop-Aware Reward** cuts bad_stop by 80% and triples training consistency with ~20 lines of monkey-patch — no framework changes.
- **Steps beat tuning.** extending to 5e7 steps gives the largest single gain (SR +4.5 pp, SPL +10.1%), confirming scene coverage as the bottleneck.
- **Sim-to-real works.** the SR 0.945 policy navigated end-to-end on WHEELTEC S100 in a real office — direct transfer without an intermediate simulator.
- **Future work.** multi-scene real-robot tests, dynamic obstacles, and LiDAR-only observations (TurtleBot3-class platforms).

10 · References

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Acknowledgments — FURP 2026 · Faculty of Science and Engineering · University of Nottingham Ningbo China. Training: NVIDIA RTX 4090D. Robot: WHEELTEC S100 (Jetson Orin Nano · Orbbec Astra S · ROS1 Noetic). Code, results and demo video: project repository.

Habitat · HM3D · PPO · Stop-Aware Reward · Sim-to-Real